

Thinking About Answers

The Application of
Cognitive Processes to
Survey Methodology

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Answering a Survey Question

Cognitive and Communicative Processes

Several researchers have offered conceptual models of the question asking and answering process. Initially these models presented straightforward extensions of general information processing models developed in cognitive psychology (for example, Bodenhausen & Wyer, 1987; Hastie, 1987), described in detail by Lachman, Lachman, and Butterfield (1979). However, as the collaboration of cognitive psychologists, social psychologists, and survey researchers progressed, it became increasingly obvious that the information processing paradigm could only capture some of the key aspects of survey interviews.

Most important, mainstream information processing models share an exclusive focus on individual thought processes. As many critics noted, this concentration on individuals as isolated information processors fostered a neglect of the social context in which human judgment occurs (Forgas, 1981; Schwarz, 1994; Schwarz & Strack, 1991c). The survey interview, however, is best considered as an ongoing conversation in which respondents conduct their own share of thinking and question answering in a specific social and conversational context. Hence, conceptualizations of the question answering process in survey interviews need to consider conversational as well as cognitive processes and pay close attention to the complex interplay of social communication and individual

thought processes (see Clark & Schober, 1992; Schwarz, 1994; Strack & Schwarz, 1992, for more detailed discussions).

In the present chapter we begin by summarizing respondents' tasks. We then discuss in some detail each of these tasks and the cognitive and communicative processes involved in their performance, providing a conceptual framework for the following chapters. Those chapters will elaborate on different aspects of the processes introduced here as they become relevant to different issues of survey research.

Overview of Respondents' Tasks

From a cognitive perspective, answering a survey question requires that respondents accomplish several tasks. There is wide agreement among researchers on the substantive nature of these tasks although different researchers use somewhat different labels (see, for example, Feldman, 1992; Feldman & Lynch, 1988; Schwarz, 1990a; Schwarz & Bless, 1992a; Strack, 1992, 1994; Strack & Martin, 1987; Tourangeau, 1984, 1987, 1992; Tourangeau & Rasinski, 1988). In the following paragraphs we discuss the various tasks as if they were sequentially ordered. This facilitates the exposition; however, respondents may in fact go back and forth between different tasks. The sequential order of our presentation should not, therefore, be taken too literally.

Not surprisingly, respondents' first task consists in interpreting the question to understand its meaning. If the question is an opinion question, they may either retrieve a previously formed opinion from memory or may "compute" an opinion on the spot. To do so they need to retrieve relevant information from memory to form a mental representation of the target they are to evaluate. In most cases, they also need to retrieve or construct some standard against which they can evaluate the target. After they form a "private" judgment they have to communicate it to the researcher. To do this, they may need to format the judgment to fit the response alter-

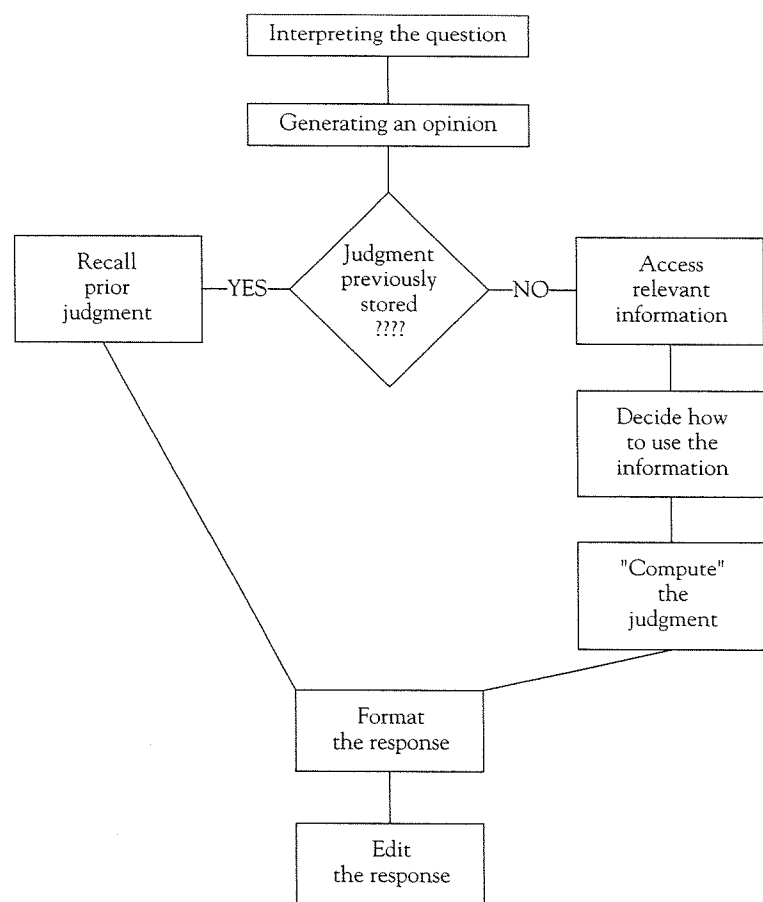
natives provided as part of the question. In addition, they may wish to edit their response before they communicate it because of social desirability and situational factors.

Similar considerations apply to behavioral questions. Again, respondents first need to understand the question and the kinds of behavior they are supposed to report. Next, they have to recall or reconstruct relevant instances of the behavior from memory. If the question specifies a reference period, they must also determine if the instances occurred during that period. Similarly, if the question refers to "usual" behavior, respondents have to determine if the recalled or reconstructed instances are reasonably representative or if they reflect a deviation from their usual behavior. If they cannot recall or reconstruct specific instances of the behavior or if they are not sufficiently motivated to engage in this effort, respondents may rely on general knowledge or other salient information to help them come up with an estimate. Finally, respondents have to provide the answer to the researcher. They may need to map the estimate onto a response scale provided to them and, again, they may want to edit it for reasons of social desirability.

Accordingly, interpreting the question, retrieving information, generating an opinion or a representation of the relevant behavior, formatting a response, and editing it are the main psychological components of a process that starts with respondents' exposure to a survey question and ends with their overt report. Figure 3.1, which we adapted from Strack and Martin's (1987) discussion of attitude questions, summarizes these steps.

Although the actual question answering sequence ends with the respondent's answer, the processes involved in answering a question may themselves change a respondent's cognitive representation of the issue and affect their subsequent behavior. Feldman and Lynch (1988) referred to this possibility as *self-generated validity*. This term means that the measurement process can sometimes generate the very thing it is supposed to assess. We address this consequence of survey participation as the final step of the question answering process.

FIGURE 3.1 Model of information processing in a survey situation.



Understanding the Question

The key issue in question comprehension is whether the respondent's understanding of the question matches what the researcher had in mind. Is the attitude object—or the behavior—that the respondent identifies as the target of the question the one that the researcher intended? Does the respondent's understanding tap the same facet of the issue and the same evaluative dimension?

From a psychological point of view, question comprehension

reflects the operation of two intertwined processes (Clark & Schober, 1992; Strack, 1992; Strack & Schwarz, 1992). The first involves semantic understanding of the utterance. If the words used are ambiguous or unfamiliar, for example, respondents need to disambiguate their meaning. However, mere understanding of the words is not sufficient. For example, if respondents are asked "What have you done today?" they are likely to understand the meaning of the words. However, they still need to determine the kinds of activities the researcher is interested in. Should they report, for example, that they took a shower? Hence, understanding a question in a way that allows an appropriate answer requires understanding not only the *literal meaning* of the question but involves inferences about the questioner's intention. To infer the questioner's intention, that is, the *pragmatic meaning* of the question, respondents rely on the tacit assumptions that govern the conduct of conversation in everyday life. Next, we consider the processes involved in question comprehension in more detail.

Understanding the Literal Meaning

Language comprehension begins with a perceptual process that transforms the sounds of an utterance or the symbols of written language into words. Next, the string of words is "parsed" into meaningful units, and the lexical meaning of the identified words is recalled from semantic memory (see Anderson, 1980; Glass, Holyoak, & Santa, 1979, for more detailed discussions). Performance of each of these steps is very context-dependent, although we are usually not aware of the degree to which we use context in apparently simple tasks of comprehension.

At the perceptual level, word recognition is relatively context-independent if the word is presented in writing. However, spoken words are more difficult to recognize and numerous studies demonstrate that a given spoken word is easier to recognize in a meaningful context than in isolation (for example, Lieberman, 1963; see Marslen-Wilson, 1984, for a review). Moreover, the same string of letters or sounds may acquire different meaning, depending on the

syntax of the sentence. Thus, the correct meaning of a word can only be identified after the sentence is parsed into meaningful syntactic units. Consider the sentence "The man pushed through the door fell" (taken from Smyth, Morris, Levy, & Ellis, 1987). Most readers will need to revise their initial interpretation when they reach the end of the sentence. As Anderson (1980, p. 413) noted, listeners "select only one interpretation (their best guess) for a pattern and carry it through to the end of the sentence. If the best guess turns out to be wrong, their comprehension suffers and they have to backtrack and try another interpretation."

Most comprehension problems arise because of either lexical or structural ambiguities. In the sentence "John went to the bank," the word "bank" is a source of lexical ambiguity because it can refer to a financial institution or a riverbed. In contrast, the sentence "Flying planes can be dangerous" is structurally ambiguous because "flying" may be interpreted as a verb or as an adjective.

Whereas structural ambiguities can be avoided by careful question wording, lexical ambiguities are inherent in language and reflect the multiple meanings of words. In most cases, these multiple meanings are constrained by the context in which words appear. If these constraints allow for several different meanings, the interpretation used is the one that comes to mind most easily. In general, individuals do not entertain different interpretations simultaneously but select the most accessible one and carry it through until potentially proven wrong.

Which of the multiple meanings of a word comes to mind most easily in a survey interview is a function of questionnaire and respondent variables. At the questionnaire level, the content of preceding questions may increase the accessibility of a concept, rendering it likely that this concept rather than another is used in resolving subsequent lexical ambiguities. For example, Lackner and Garrett (1972) asked subjects to interpret the sentence "The spy put out the torch as our signal to attack." In this case, "put out" could be interpreted as either "switched off" or "made visible." Subjects were more likely to choose the former meaning when a simul-

taneous second task primed the concept "extinguished" and the latter meaning when it primed the concept "displayed." When asked, however, subjects were not aware of any influence of the accompanying second task, illustrating the unconscious and automatic nature of semantic priming effects. However, not all question order effects at the comprehension stage reflect such an automatic encoding process. Rather, the impact of preceding questions may also reflect conscious inferences about intended meaning, addressed in the following section.

Independent of question context, different meanings may be differentially accessible to different respondents because of the frequency with which they employ them in daily life. In an amusing example, Billiet (cited in Bradburn, 1992, p. 317) observed that some respondents offered numbers between twenty and thirty in response to the question "How many children do you have?" Closer inspection revealed that these respondents were teachers who interpreted the question to refer to the children in their classes, the meaning that was most accessible in their memories.

The problem of differential lexical meanings is further compounded by the existence of idiosyncratic meanings and by differences in meanings across regions and subcultures. In addition, many of the terms used in public opinion research do not have clearly defined lexical meanings to begin with and respondents who ask the interviewer to provide a definition are usually instructed to define the concept for themselves. As a result, it often remains unclear what a term meant to a respondent. A study by Fee (1979) nicely illustrates these ambiguities. Exploring the meaning of common political terms, Fee observed at least nine different meanings for the term "energy crisis." Similarly, the term "big government" elicited at least four distinct representations: one referred to welfare and overspending, another to big business and government for the wealthy, a third to federal control and diminished states' rights, and a fourth to bureaucracy and a lack of democratic process. Needless to say, it is nearly impossible to interpret the responses to a question about "big government" without knowing which interpretation the

respondents chose. Finally, the meaning of a term may change over time, posing considerable problems for trend analyses.

Not surprisingly, these problems are compounded when we want to compare survey data from different countries. In designing comparative surveys, researchers usually try to word questions in the same way across languages, using iterations of translation into and back out of other languages (see Brislin, 1980; Paarek & Rao, 1980, for discussions of cross-cultural surveys). To the extent that similar terms have different meanings in different countries, however, different questions may be needed to convey a single meaning.

In summary, the comprehension of the literal meaning of an utterance involves the identification of words, the recall of lexical information from semantic memory, and the construction of a meaning of the utterance, which is constrained by its context. However, understanding the literal meaning of a question is often not sufficient to answer it. As cooperative participants in a conversation, we also need to identify the intended meaning of the question to determine the information the questioner actually desires.

Inferring the Intended Meaning: The Logic of Conversation

To understand how respondents infer the intended meaning of a question, we need to consider the assumptions that govern the conduct of conversation in everyday life. These tacit assumptions were systematically described by Paul Grice (1975), a philosopher of language (see Levinson, 1983, for a detailed introduction; Sperber & Wilson, 1986, for a recent extension; and Clark & Schober, 1992; Schwarz, 1994; Schwarz & Hippler, 1991; Strack, 1994; Strack & Schwarz, 1992, for applications to survey research).

According to Grice's analysis, conversations proceed according to a "cooperativeness" principle. This principle can be expressed in the form of four maxims. A *maxim of quality* enjoins speakers not to say anything they believe to be false or lack adequate evidence for. A *maxim of relation* enjoins speakers to make a contribution relevant

to the aims of the ongoing conversation. A *maxim of quantity* requires speakers to make their contribution as informative as required but not more informative than required, and a *maxim of manner* holds that the contribution be clear rather than obscure, ambiguous, or wordy. Thus, according to Grice, speakers are supposed to be truthful, relevant, informative, and clear.

As a result, "communicated information comes with a guarantee of relevance" (Sperber & Wilson, 1986, p. vi) and listeners interpret the speakers' utterances "on the assumption that they are trying to live up to these ideals" (Clark & Clark, 1977, p. 122). These tacit assumptions have important implications for survey research, implications to which we shall repeatedly return throughout this book.

To begin with an extreme example, the assumption that speakers try to be truthful, relevant, informative, and clear implies that they are unlikely to ask meaningless questions. If a survey researcher does so nevertheless, as is the case with questions about highly obscure (for example, Schuman & Presser, 1981) or completely fictitious issues (for example, Bishop, Oldendick, & Tuchfarber, 1986), respondents are unlikely to assume that the researchers violated each and every principle that usually governs the conduct of conversation. Hence, they will operate on the default assumption that the question is meaningful and do their best to determine what the researchers might have had in mind when they wrote it. In doing so, respondents turn to the context of the question and the response alternatives for relevant clues. This strategy is licensed by the maxims of conversation.

Even if comprehending the literal meaning of the question poses no problem, respondents still have to infer what the specific information is that the researchers are interested in. Speakers are supposed to provide information that the recipient needs rather than reiterate information that the recipient already has—or may take for granted anyway (Haviland & Clark, 1974). Accordingly, determining the intended meaning of a question and the kind of response that is adequate requires a considerable degree of inference,

including inferences about the information that the recipient already has and the kind of additional “new” information that the recipient may need. In making these inferences, respondents not only rely on the wording of the question but also consider the response alternatives, the context in which the question is presented, and their own previous answers.

Response Alternatives

Suppose, for example, that respondents are asked in an open response format, “What have you done today?” To give a meaningful answer, respondents must determine which activities are of interest to the researcher. In an attempt to be informative, respondents are likely to omit activities that the researcher is already aware of, such as “I participated in a survey interview,” or probably takes for granted, such as “I took a shower.” But if respondents are given a list of activities that includes being interviewed and taking a shower, most respondents will endorse them. However, presenting such a list would reduce the likelihood that respondents report activities that are not represented on it (see Schuman & Presser, 1981; Schwarz & Hippler, 1991, for a review of relevant studies). Both of these question-form effects reflect the fact that response alternatives can clarify the intended meaning of a question, in the present example by specifying the activities the researcher is interested in. Whereas this example may seem rather obvious, more subtle influences are frequently overlooked.

Suppose that respondents are asked how frequently they felt “really irritated” recently. To answer this question, they again must determine what the researcher means by “really irritated.” Does the term refer to major or to minor annoyances? To identify the intended meaning, they may consult the frequency response alternatives provided by the researcher. If the response alternatives offer low-frequency categories (for example, ranging from “less than once a year” to “more than once a month”) they may conclude that the researcher has relatively rare events in mind and that the question

does not refer to minor irritations, which are likely to occur more often. Accordingly, Schwarz, Strack, Müller, and Chassein (1988) observed that respondents who were asked to report the frequency of irritating experiences on a low-frequency scale assumed that the question referred to major annoyances whereas respondents who had to give their report on a high-frequency scale assumed that the question referred to minor annoyances. Thus, respondents inferred a different meaning for the question based on the response alternatives provided.

Similarly, Schwarz, Knäuper, Hippler, Noelle-Neumann, and Clark (1991) observed that respondents may use the specific numeric values provided as part of a rating scale to interpret the meaning of the scale’s labels. In this study, a representative sample of German adults was asked, “How successful would you say you have been in life?” This question was accompanied by an 11-point rating scale, ranging from “not at all successful” to “extremely successful” However, in one condition the numeric values of the rating scale ranged from 0 (“not at all successful”) to 10 (“extremely successful”), whereas in the other condition they ranged from –5 (“not at all successful”) to +5 (“extremely successful”). The results showed a dramatic impact of the numeric values presented to respondents. Whereas 34 percent of the respondents endorsed a value between 0 and 5 on the 0 to 10 scale, only 13 percent endorsed one of the formally equivalent values between –5 and 0 on the –5 to +5 scale. Subsequent experiments indicated that the results reflected differential interpretations of the term “not at all successful.” When this term was combined with the numeric value 0, respondents interpreted it to reflect the absence of success. However, when the same term was combined with the numeric value –5 and the scale offered a 0 as the midpoint, they interpreted it to reflect the presence of failure.

This differential interpretation of the same term as a function of its accompanying numeric was also reflected in the inferences made by judges based on a report given along a rating scale. For example, in one experiment a fictitious student reported his academic

success along one of the above scales, checking either a -4 or a 2. Judges who were asked to estimate how often this student had failed an exam assumed that he had failed twice as often when he checked a -4 than when he checked a 2, although both values are formally equivalent along 11-point rating scales of the type described.

Extending this line of work, Grayson, Schwarz, and Hippler (1995) asked respondents to provide behavioral frequency reports on a rating scale ranging from "rarely" to "frequently." Depending on experimental conditions, the verbal label "rarely" was coded with the numeric value 0 or the numeric value 1. As expected, respondents provided higher frequency ratings when "rarely" was coded 0 rather than 1. The numeric value 0 induced them to interpret the verbal label "rarely" as meaning "never," thus requiring the assignment of higher numbers for a given behavioral frequency.

In combination, such findings demonstrate that respondents use the response alternatives in interpreting the meaning of a question. In doing so, they proceed on the tacit assumption that every contribution is relevant to the aims of an ongoing conversation. In the survey interview, these contributions include apparently formal features of questionnaire design, such as the numeric values given on a rating scale. Hence, identically worded questions may acquire different meanings depending on the response alternatives by which they are accompanied (see Schwarz & Hippler, 1991, for a more extended discussion).

Question Context

Respondents' interpretation of a question's intended meaning is further influenced by the context in which the question is presented. Not surprisingly, this influence is more pronounced as the wording of the question becomes more ambiguous. From a theoretical point of view, some contextual influences may reflect the operation of semantic priming, as discussed earlier, as well as the operation of conscious inference processes on the basis of conversational assumptions.

For example, Strack, Schwarz, and Wänke (1991, Experiment 1) asked German students to report their attitude toward an ambiguous issue: the introduction of an undefined "educational contribution." For some respondents, this question was preceded by a question about the tuition students have to pay in the United States; for others it was preceded by a question about the financial support Swedish students receive from their government. As expected, respondents reported a more favorable attitude toward the educational contribution when the preceding question referred to money students receive rather than to money they have to pay. (It should be noted that university education is free in Germany.) This finding may indicate that the preceding questions primed different concepts that were subsequently used automatically in the encoding of the ambiguous question or that the respondents deliberately searched the context of the ambiguous question to make sense of it. Although we cannot distinguish these processes in the Strack, Schwarz, and Wänke study, other studies have demonstrated that questions that follow rather than precede an ambiguous question may result in similar disambiguations in self-administered questionnaires (see Schwarz, Strack, Hippler, & Bishop, 1991). This suggests that respondents deliberately search the context of the ambiguous question to make sense of it, even to the extent of returning to the question after they encounter a subsequent one that allows a meaningful interpretation. How often such backtracking occurs in self-administered questionnaires is unknown. We assume that it is most likely to occur when respondents encounter a question that does not make sense to them until a subsequent question allows for a reasonable interpretation.

Moreover, a question's context may influence its interpretation even under conditions where the literal meaning of the question does not seem especially ambiguous. Two examples may illustrate this point. Schwarz, Strack, and Mai (1991) asked respondents to report their general life satisfaction, which did not pose a particular comprehension problem. However, in one condition of their experiment, this question was preceded by a question about the respondent's marital satisfaction and both questions were introduced by a

joint lead-in. This lead-in read, "We would first like to ask you to report on two aspects of your life that may be relevant to people's overall well-being." In this case, respondents interpreted the general life satisfaction question as if it were worded, "Leaving aside the life domain that you already told us about, how satisfied are you currently with other aspects of your life?" This shift in the interpretation of the general life satisfaction question reflects the impact of the "given-new" contract (Haviland & Clark, 1974) that requests that participants provide information that is new to the recipient rather than to reiterate information already given. Accordingly, respondents deliberately excluded their marriage in answering the general question because they had reported on it earlier. (We return to this study in more detail in Chapters Four and Five.) In a similar vein, Bachman and Alcser (1993) asked respondents to report their satisfaction "with the current U.S. health care system" and their satisfaction with their own health insurance plans. Most respondents who had health insurance reported high satisfaction with their insurance plans, independently of the order in which both questions were presented (77.8 percent were very or somewhat satisfied when the question was asked first and 76.4 percent when it was asked second). Their reported satisfaction with the U.S. health care system in general, however, showed a pronounced order effect. When the question about the U.S. health care system was asked first, 39.6 percent of the respondents reported being very or somewhat satisfied whereas only 26.4 percent did so when this question was preceded by the question about their own insurance plans. This pattern of findings suggests that respondents who had just reported on their own health insurance plans, with which most were satisfied, excluded this information from consideration when they were asked to evaluate the U.S. health care system in general. Having already provided information about their own situation, they apparently interpreted the general question as if it were worded, "Aside from your own health insurance, how satisfied are you with the U.S. health care system in general?" resulting in reports of lower satisfaction.

As these findings illustrate, question comprehension is not merely a function of the wording of the question itself. Respondents use information provided by the context of the question to determine its intended meaning. In making these pragmatic inferences, they rely on the principles that govern the conduct of conversation in daily life. Accordingly, models of the question comprehension process must include pragmatic as well as semantic processes (see Clark & Schober, 1992; Schwarz, 1994; Strack, 1994; Strack & Schwarz, 1992, for more detailed treatments).

Recalling or Computing a Judgment

Once respondents determine what the researcher is interested in, they need to recall relevant information from memory. In some cases, respondents may have direct access to a previously formed relevant judgment that they can offer as an answer. In most cases, however, they will not find an appropriate answer readily stored in memory and will need to compute a judgment on the spot.

Recalling a Judgment

Whether respondents can recall a previously formed relevant judgment from memory depends on whether such a judgment has been formed in the first place and whether it is accessible at the time of the interview. In the case of attitude questions, a key determinant is the personal importance of the issue and the degree of personal experience with the attitude object. Not surprisingly, issues of personal importance are more likely to elicit spontaneous judgments than are less important ones. Moreover, some daily activities, such as major purchasing decisions, require the evaluation of different objects, and if the decision was made recently the evaluations formed at that point may still be accessible in memory. In addition, the likelihood that a respondent has access to evaluative judgments of an attitude object increases with the degree of the respondent's personal experience with the object. (See Fazio, 1989, for a review

of attitude accessibility research, and Bassili, 1995, for an application to survey research.) Finally, if respondents have been asked a related question before, the judgment formed at that time may still be accessible in memory, provided that little interfering information has been activated in the meantime.

In the case of behavioral questions, a relevant answer is most likely to be directly accessible if the behavior is of personal importance and has a low frequency of occurrence, as we shall see in the chapters on behavioral reports (see Bradburn, Rips, & Shevell, 1987; Schwarz, 1990a; Strube, 1987, for reviews). If the behavior is a frequent one, respondents are only likely to have direct access to a judgment if the behavior is highly regular, in which case they may remember a rate of occurrence, such as "once a week" (see Menon, 1994).

Computing a Judgment

Usually, respondents need to compute a judgment when asked a question. In the case of attitude questions, this is because issues are complex whereas survey questions are necessarily simple, as Schuman and Kalton (1985) noted. Thus, even under conditions where respondents can retrieve an opinion on the issue from memory, the opinion may not exactly match the facet tapped in the question. Similarly, respondents are unlikely to have an appropriate answer to most behavioral questions stored in memory. Even if they can recall relevant instances, they will still need to determine if these instances fit the reference period, and so on. As a result, most of the answers that we record in surveys reflect judgments that respondents generate on the spot in the specific context of the specific interview. They are therefore strongly influenced by the information that is accessible at that time and this is in part a function of the preceding questions.

This context-dependency of human judgment results from the fact that humans are "cognitive misers" (Taylor, 1981), who follow a "satisficing" rather than an "optimizing" strategy (see Simon,

1957, for an introduction to these concepts, and Krosnick, 1991, for their application to survey responding). When asked to form a judgment, individuals first need to form a mental representation of the target to which the question pertains, as we shall discuss in more detail in Chapter Five. To do so, they do not retrieve all potentially relevant information. Rather, they truncate the search process as soon as enough information has come to mind to form a representation that is sufficient for the judgment at hand. (See Bodenhausen & Wyer, 1987, for a more detailed discussion.) An extensive search of memory is only likely when individuals are highly motivated because the judgment has important personal consequences (Kruglanski, 1980). This is rarely the case in survey interviews. Moreover, searching memory takes time and the time pressures that characterize most survey interviews (see Groves & Kahn, 1979) further contribute to individuals' tendency to truncate the search process early. Accordingly, respondents' judgments are usually based on the information that comes to mind most easily while other potentially relevant information that is less accessible in memory is unlikely to be considered. How easily a given piece of information comes to mind depends on the recency and frequency of its use and the degree to which it is linked to other pieces of information (see Bodenhausen & Wyer, 1987; Higgins, 1989; Schwarz & Sudman, 1995; Wyer & Srull, 1989, for reviews).

Information that has just been used to answer the preceding question is particularly likely to come to mind when a related question is asked. The information has been rendered "temporarily accessible" by the preceding question and is likely to serve as a basis for later judgments to which it may be relevant. This priming function of preceding questions is the procedural basis of many context effects in survey measurement, which we consider in more detail in Chapters Four and Five. In addition to the recency of its use, the accessibility of a given piece of information is further influenced by the frequency of its use. For example, a respondent who is concerned about the threat of unemployment may frequently think about this issue, rendering it "chronically accessible" in memory. As

a result, this respondent may think of unemployment spontaneously if a relevant political question is asked whereas other respondents may think of unemployment only if it was addressed in a preceding question. As we have already discussed in the context of lexical meanings, the *temporary accessibility* of information is largely a function of questionnaire variables, whereas the *chronic accessibility* of information is largely a function of respondent variables.

Finally, a given piece of information is proportionately more accessible in memory the more it is integrated into a larger knowledge structure. Most cognitive psychologists portray memory as an extended network, in which individual pieces of information are represented as nodes (see Anderson, 1980, for a textbook length introduction, or Hastie, 1987, for a short summary). A search of memory proceeds along the pathways that connect these nodes, and a given node is more likely to be accessed if more pathways lead to it. In terms of the previous example, a respondent who is concerned about unemployment may have thought about this problem with regard to many aspects of his personal life, thus establishing many pathways. As a result, unemployment may come to mind when any of the aspects to which it has been linked is activated.

Whereas the accessibility of information in memory determines *which* information is used in forming a judgment, what exactly respondents do with the information that comes to mind depends on the inferential procedures that they apply to it. Not surprisingly, the procedures that are used to generate an opinion or to generate a behavioral report differ substantially, and we shall explore these procedures in the respective chapters. For the time being, we note that survey responses are based on the information that is most accessible at the time of judgment. Respondents use this information to form a mental representation of the attitude object or of the relevant behavior and base their reports on the implications of this representation. Some of the information that enters this representation is chronically accessible in memory and likely to come to mind whenever the respondent thinks of the issue. This chronically accessible information contributes to the stability of reports at dif-

ferent points in time and in different contexts. Other information, however, is only temporarily accessible, for example, because it was used to answer a preceding question. This information contributes to the instability of reports over time and underlies the emergence of context effects in survey measurement. We shall repeatedly return to these issues throughout this book.

Formatting the Response

Once respondents have formed a judgment, the interviewer usually does not allow them to report it in their own words. Rather, they are asked to endorse one of the response alternatives provided. In order to do this, they must format their response in line with the options given. Accordingly, the researcher's choice of response alternatives may strongly affect survey results (see Schwarz & Hippler, 1991, for a review).

From a theoretical point of view, however, it is important to note that the influence of response alternatives is not limited to the formatting stage and that response alternatives are likely to influence all steps of the question answering sequence. As an illustration, consider the case of categorical scales that offer a number of discrete opinions or behaviors, of which the respondent is supposed to check the appropriate one. If respondents cannot identify a response alternative that reflects their judgment, they are unlikely to report it, even if a generic "other" category is offered for such cases (for example, Schuman & Presser, 1981). This well-documented finding, for example, may reflect constraints at the formatting stage as well as differences in respondents' inferences about the intended meaning of the question, as discussed previously. Similarly, the range of the frequency categories that accompany a behavioral question may influence question comprehension (for example, Schwarz, Strack, Müller, & Chassein, 1988) as well as respondents' judgmental strategy (for example, Schwarz, Hippler, Deutsch, & Strack, 1985), in addition to the formatting process.

The only scale effects that seem to occur unequivocally at the

formatting stage pertain to the use of rating scales, as described by Ostrom and Upshaw's (1968) perspective theory and Parducci's (1983; Parducci & Perrett, 1971) range-frequency theory. As numerous studies demonstrate, respondents use the most extreme stimuli to anchor the endpoints of a rating scale. As a result, a given stimulus will be rated as less extreme if presented in the context of a more extreme stimulus than if presented in the context of a less extreme one. In Parducci's model, this impact of the range of stimuli is referred to as the "range effect." In addition, if the number of stimuli to be rated is sufficiently large, respondents attempt to use all categories of the rating scale about equally often. Accordingly, the specific ratings given also depend on the frequency distribution of the presented stimuli, an effect that is referred to as the "frequency effect." Daamen and de Bie (1992) provide an introduction to the logic of these processes and report several studies that illustrate their impact on survey results. We return to these issues in Chapter Four.

Editing the Response

Finally, respondents may want to edit their responses before they communicate them, reflecting considerations of social desirability and self-presentation. Not surprisingly, the impact of these considerations is more pronounced in face-to-face interviews than in self-administered questionnaires (for example, Smith, 1979). DeMaio (1984) reviews the survey literature on this topic.

It is important to emphasize, however, that the size of response effects reflecting influences of social desirability is usually modest and these effects are limited to potentially threatening questions. Moreover, what constitutes a socially desirable response depends on the situation. For example, several researchers (see Smith, 1979, for a review) observed that respondents report higher levels of happiness and satisfaction in face-to-face interviews than in self-administered questionnaires. In contrast, Strack, Schwarz, Chassein, Kern, and Wagner (1990) obtained a reversal of this effect

under specific conditions. In their study, respondents reported deflated levels of happiness when they were questioned by a disabled interviewer, presumably because it seemed inappropriate to tell an unfortunate other how wonderful their own lives were. In contrast, the sheer presence of a disabled confederate while respondents filled out a self-administered questionnaire resulted in increased reports of happiness, indicating that the disabled person served as a salient standard of comparison inflating the respondents' private judgments. As this example illustrates, understanding issues of social desirability requires close attention to the actual social situation, which determines what is desirable and what is not.

Self-Generated Validity: Consequences of Survey Participation

Although the question answering sequence ends with the respondent's report, it is useful to consider the impact of the measurement process on the respondent's subsequent attitudes, intentions, and behaviors. Feldman and Lynch (1988; see also Feldman, 1992) provide a detailed discussion of the underlying processes, which we shall address in more detail in our discussion of context effects in attitude measurement. In principle, answering a question about one's behavioral intentions increases the accessibility of these intentions in memory and hence the likelihood that they may come to mind in a situation in which one may act on them.

Sherman and his colleagues (Sherman, 1980; Sherman, Zehner, Johnson, & Hirt, 1983) observed that the very act of making a prediction about one's future behavior can induce behavioral change that brings the future behavior in line with one's predictions. In one of their experiments, college students were asked to predict whether they would volunteer three hours for a cancer society's fundraising campaign. These subjects typically overestimated their willingness relative to the actual behavior of a control group that was simply asked to volunteer three hours. But these errors were "self-erasing" in the sense that subjects who made a prediction subsequently acted

on the basis of their predictions, resulting in a higher level of actual volunteer behavior. Similarly, Gregory, Cialdini, and Carpenter (1982) found that consumers who were induced to consider a scenario in which they were cable television subscribers subsequently were more likely actually to subscribe to cable television service. Moreover, Greenwald, Carnot, Beach, and Young (1987) demonstrated that asking respondents whether they intended to vote increased their actual participation in an election.

At present, the most dramatic evidence of the impact of survey measurement on subsequent behavior has been reported by Morwitz, Johnson, and Schmittlein (1993). Using data from a consumer panel survey with more than forty thousand households, the authors observed that being asked a question about one's intention to buy a car or a personal computer increased the likelihood of actual purchase. Whereas 2.4 percent of the households surveyed in that study bought a car within a six-month time period when they were not asked an intention question in the previous wave, 3.3 percent of the households did so when they were asked an intention question. This reflects an increase of 37 percent over the baseline of 2.4 percent as a result of being asked a single question, namely, "When will the next new car be purchased by someone in your household?" Similarly, 3.8 percent of the households reported the purchase of a personal computer without having been asked an intention question, whereas 4.5 percent did after having been asked in a preceding wave, "Do you or does anyone in your household plan to acquire a (another) personal computer in the future for use at home?" This reflects an increase of 18 percent over the baseline. Thus, "the proportion of households that acquired a product was greater for households whose intentions were measured once than for those whose intentions were never measured" (Morwitz, Johnson, & Schmittlein, 1993, p. 52). Additional analyses indicated that the impact of measurement was more pronounced for households that had no prior experience with the respective product than for households that had.

Finally, repeated intent measures over several waves resulted in

a polarization of respondents' intentions, increasing the intentions of respondents who initially reported a high intent and significantly decreasing the intentions of those who initially reported a low intent. These analyses also revealed that repeated intent questions increased actual purchases for households with a high initial purchase intent but decreased purchases for those with a low initial intent, in particular those with no prior product experience. Thus, thinking about purchase intent in response to survey questions increased the likelihood of corresponding actual behavior.

In summary, these studies illustrate that answering survey questions may influence not only responses to subsequent questions but also the actual behavior of respondents. The data suggest that such behavioral effects, if they occur, confirm the response provided in the survey, resulting in what Feldman and Lynch (1988) have termed "self-generated validity."

Summary

In this chapter we discussed what respondents must do to answer a question. For ease of exposition, we presented respondents' tasks in a sequential order. We assume that this sequential order is, in fact, a correct portrayal of the sequence respondents go through in many cases. But it is equally plausible that respondents may switch back and forth between these tasks. They may notice, for example, that their initial interpretation of a question was not the one the researcher intended when they try to report their answer and cannot find an appropriate response alternative. Or they may realize that the researcher is interested in a more detailed behavioral report than they initially assumed when they encounter highly detailed response alternatives that may prompt them to engage in more detailed recall efforts. Hence, thinking of these tasks as sequentially ordered is plausible but may not always capture respondents' actual actions. Nevertheless, the sequential ordering of the tasks provides a useful framework for our purposes.

To reiterate, respondents first have to interpret the question to understand what is meant. If the question is an opinion question, they may retrieve a previously formed opinion from memory or “compute” an opinion on the spot. To do so, they need to retrieve relevant information from memory to form a mental representation of the target that they are to evaluate. In most cases, they will also need to retrieve or construct some standard against which the target is evaluated. Once respondents form a “private” judgment, they have to communicate it to the researcher. To do so, they may need to format the judgment to fit the response alternatives provided or they may wish to edit their responses before communicating them, because of influences of social desirability and situational factors.

Similar considerations apply to behavioral questions. Again, respondents first need to understand what the question refers to and which behavior they are supposed to report. Next, they have to recall or reconstruct relevant instances of this behavior from memory. If the question specifies a reference period, they must also determine if these instances occurred during the reference period or not. Similarly, if the question refers to “usual” behavior, respondents have to determine if the recalled or reconstructed instances are reasonably representative or if they reflect a deviation from their usual behavior. If they cannot recall or reconstruct specific instances of the behavior or are not sufficiently motivated to engage in this effort, respondents may rely on their general knowledge or other salient information that bears on the task to compute an estimate. Finally, respondents have to provide their estimate to the researcher. They may need to map it onto a response scale provided and they may want to edit it for reasons of social desirability.

To understand how respondents infer the intended meaning of a question, we need to consider communicative as well as cognitive processes. The communicative processes reflect the fact that respondents bring the assumptions that govern the conduct of conversation in everyday life to the survey interview. These tacit assumptions were systematically described by Paul Grice (1975).

Drawing on these assumptions, respondents use cues provided by the context of the question and the response alternatives to determine the question’s intended meaning. In addition, the content of preceding questions increases the cognitive accessibility of information that was used to answer them. It therefore affects what is likely to come to mind later on and influences which information respondents draw on in answering subsequent related questions.

Finally, answers to survey questions not only influence other answers in the survey but also may influence future behavior.